Tutorial 2: Maximum Likelihood and Least Squares

Likelihood interpretation, estimation, and regression

Kangning Cui
City University of Hong Kong (Dongguan)

Semester A, 2026/27

Tutorial 2: from likelihood to learning objectives

This tutorial reinforces the main ideas from Week 2:

- interpret likelihood correctly;
- understand Gaussian variance estimation;
- connect least squares with its geometric meaning;
- understand how probability assumptions create different losses.

Task 1: likelihood interpretation and boundary cases

Bernoulli model ▪ 12 minutes

A coin follows a Bernoulli model:

$$p(x \mid \pi) = \pi^x (1-\pi)^{1-x}, \qquad x \in \{0, 1\}.$$

Consider two datasets:

$$A = (1, 1, 1, 1, 1), \qquad B = (1, 0, 1, 1, 0).$$

Questions

1. In a likelihood function, what is fixed and what changes?
2. Derive the MLE of π for datasets A and B.
3. Why does dataset A produce a boundary solution?

Solution 1: likelihood interpretation

In likelihood, observations are fixed and π varies.
For Bernoulli data,

$$L(\pi) = \prod_i \pi^{x_i}(1-\pi)^{1-x_i} = \pi^m(1-\pi)^{n-m},$$

where m is the number of heads.
For dataset $A = (1, 1, 1, 1, 1)$:

$$m = 5, \qquad n = 5, \qquad L(\pi) = \pi^5.$$

Since $L(\pi)$ increases as π approaches 1,

$$\hat{\pi}_{\mathrm{ML}} = 1.$$

Solution 1: constructing the likelihood

For dataset $B = (1, 0, 1, 1, 0)$:

$$m = 3, \qquad n = 5.$$

Therefore,

$$L(\pi) = \pi^3 (1 - \pi)^2.$$

Maximizing the likelihood is equivalent to maximizing the log-likelihood:

$$\ell(\pi) = 3 \log \pi + 2 \log(1 - \pi).$$

Taking logarithms changes products into sums, which makes differentiation easier.

Solution 1: solving the interior optimum

Differentiate the log-likelihood:

$$\frac{d\ell}{d\pi} = \frac{3}{\pi} - \frac{2}{1-\pi}.$$

At an interior optimum,

$$\frac{3}{\pi} - \frac{2}{1-\pi} = 0.$$

Therefore,

$$3(1-\pi) = 2\pi,$$

giving

$$\boxed{\hat{\pi}_{\mathrm{ML}} = \frac{3}{5}}.$$

Task 2: Gaussian variance estimation

MLE and interpretation ▪ 15 minutes

Suppose

$$x_1, \ldots, x_n \sim \mathcal{N}(\mu, \sigma^2),$$

and the mean has already been estimated:

$$\hat{\mu} = \bar{x}.$$

Questions

1. Derive the MLE estimate of σ^2.
2. Explain what the residual term $(x_i - \bar{x})^2$ represents.
3. Explain why the unbiased sample variance uses $n - 1$ instead of n.

Solution 2: Gaussian variance MLE

The Gaussian log-likelihood with $\mu = \bar{x}$ is

$$\ell(\sigma^2) = -\frac{n}{2}\log\sigma^2 - \frac{1}{2\sigma^2}\sum_i (x_i - \bar{x})^2 + C.$$

Let $v = \sigma^2$. Then

$$\frac{\partial \ell}{\partial v} = -\frac{n}{2v} + \frac{1}{2v^2}\sum_i (x_i - \bar{x})^2.$$

Setting the derivative to zero:

$$-nv + \sum_i (x_i - \bar{x})^2 = 0.$$

Therefore,

$$\boxed{\hat{\sigma}^2_{\mathrm{ML}} = \frac{1}{n}\sum_i (x_i - \bar{x})^2}.$$

Why does variance use $n-1$ instead of n?

The difference appears because the mean is estimated from the same data.
If the true mean μ is known, residuals are

$$x_i - \mu.$$

After replacing μ with $\bar{x}$, the residuals satisfy

$$\sum_i (x_i - \bar{x}) = 0.$$

This constraint means one direction of variation is no longer free.

Estimating the mean uses one degree of freedom, leaving $n-1$ independent directions for measuring variation.

Solution 2: interpreting n and $n-1$

The MLE variance uses

$$\frac{1}{n}\sum_i (x_i - \bar{x})^2.$$

The commonly used sample variance uses

$$\frac{1}{n-1}\sum_i (x_i - \bar{x})^2.$$

The difference comes from the estimation goal:

- MLE chooses the parameter that maximizes likelihood;
- unbiased estimation corrects the systematic downward bias.

Therefore, the two formulas answer different questions.

n is the MLE choice; $n-1$ is the unbiased-estimation correction.

Task 3: least squares interpretation

Geometry and model understanding ▪ 15 minutes

Consider the least-squares solution $\hat{\boldsymbol{w}}$:

$$\hat{\boldsymbol{w}} = \arg\min_{\boldsymbol{w}} \|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{w}\|_2^2.$$

Questions

1. What does the residual $\boldsymbol{r} = \boldsymbol{y} - \boldsymbol{X}\hat{\boldsymbol{w}}$ represent?
2. The solution satisfies
$$\boldsymbol{X}^\top \boldsymbol{r} = 0.$$
Explain this condition geometrically.
3. Why is the explicit inverse $(\boldsymbol{X}^\top \boldsymbol{X})^{-1}$ usually avoided in practice?
4. If two feature columns are almost identical, what problem may occur?

Solution 3: understanding the residual

The residual is

$$\boldsymbol{r} = \boldsymbol{y} - \boldsymbol{X}\hat{\boldsymbol{w}}.$$

It measures the difference between the observed targets and the predictions produced by the fitted model.
The condition

$$\boldsymbol{X}^{\top}\boldsymbol{r} = 0$$

means that the residual has zero inner product with every column of $\boldsymbol{X}$.
Therefore, no feature direction can further reduce the residual.

Solution 3: geometry and numerical issues

The closed-form expression is

$$\hat{\boldsymbol{w}} = (\boldsymbol{X}^\top \boldsymbol{X})^{-1} \boldsymbol{X}^\top \boldsymbol{y}.$$

Explicit inversion is usually avoided because numerical issues may occur.
If two feature columns are almost identical, the model cannot clearly distinguish their individual contributions.
As a result:

- different parameter values may give similar predictions;
- estimated weights become unstable;
- $\boldsymbol{X}^\top \boldsymbol{X}$ becomes ill-conditioned.

QR factorization and SVD provide more stable numerical solutions.

Optional challenge: changing the noise model

From probability assumptions to loss functions

In Gaussian regression, the noise assumption leads to squared error.
Now suppose the regression noise follows a Laplace distribution:

$$p(y_i \mid \boldsymbol{x}_i, \boldsymbol{w}) = \frac{1}{2b} \exp\left(-\frac{|y_i - \boldsymbol{w}^\top \boldsymbol{x}_i|}{b}\right).$$

Questions

1. Derive the negative log-likelihood objective.
2. Compare it with Gaussian least squares.
3. Explain why the choice of noise model affects outliers.

Solution: Laplace noise leads to absolute error

Taking negative logarithms:

$$-\ell(\boldsymbol{w}) = -\sum_i \log p(y_i \mid \boldsymbol{x}_i, \boldsymbol{w}) = n \log(2b) + \frac{1}{b} \sum_i |y_i - \boldsymbol{w}^\mathsf{T} \boldsymbol{x}_i|.$$

The first term is constant with respect to $\boldsymbol{w}$, and $1/b$ only changes the scale. Therefore,

$$\boxed{\hat{\boldsymbol{w}} = \arg\min_{\boldsymbol{w}} \sum_i |y_i - \boldsymbol{w}^\mathsf{T} \boldsymbol{x}_i|}.$$

Laplace noise leads to absolute-error regression (L1 loss).

Solution: Gaussian versus Laplace noise

Both models measure prediction residuals differently:

$$r = y - \boldsymbol{w}^{\mathsf{T}}\boldsymbol{x}.$$

Gaussian noise gives:

$$r^2$$

Laplace noise gives:

$$|r|.$$

For a large error, for example $r = 10$:

$$r^2 = 100, \qquad |r| = 10.$$

Therefore, squared error amplifies large errors much more strongly.

Tutorial 2 complete

Questions and discussion